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**Black Hole Thermodynamics:
The First and Second Laws Following Bekenstein (1973)**

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(draft one)

A summer reading project

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Abstract

We explored the extension of the laws of thermodynamics to black holes, deriving black hole thermodynamics, from a relativistic foundations. We started with special and general relativity, building up the required tensor formalism to tackle curvature and spacetime metrics needed to describe the spacetime geometry. With the mathematical framework in place we studied Schwarzschild, Reissner-Nordström and Kerr solutions, exploring their metrics and properties. We then reviewed the classical first and second law of thermodynamics and defined Shannon entropy, establishing the analogies we will use for the black hole mechanics. Following Bernstein's 1973 paper [1], we derived the first and second laws of black hole mechanics, drawing parallels to thermodynamics identifying the link between horizon area with thermodynamics entropy. We defined the Bekenstein temperature of a black hole, proportional to the surface gravity, Θ . Finally, we formulated the generalised the second law, which includes the entropy of a black hole together with the universe, ensuring there is no violation when a particle enters.

1 Relativity

In the 20th century Einstein introduced the language of differential geometry to describe gravity, moving away from Newtonian ideas of forces acting between masses. Relativity describes gravity as the curvature of space time itself, where this geometry is encoded in the metric $g_{\mu\nu}$. Distances are then described by contracting vectors dx^μ with the metric to give us the line element ds^2

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

Different physical systems correspond to different space times and different metrics, the simplest of which is seen in special relativity.

1.1 Special Relativity

Concisely *special relativity* asks what happens to space time near the speed of light?

Special relativity demands two postulates,

1. The laws of physics are invariant in all inertial reference frames.
2. The speed of light is a constant in a vacuum

The consequences of these postulates lead to many paradoxical ideas around time dilation and length contraction, however for us what's important is that now we introduce a four dimensional space time, treating temporal coordinates on equal footing as spatial ones. This geometry is still flat as we aren't looking at mass & energy yet, this is encoded by the Minkowski metric $\eta_{\mu\nu}$,

$$\eta_{\mu\nu} = \begin{bmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

Now taking a 4-vector

$$dx^\mu = \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

Our space time is then described by

$$ds^2 = dx_\mu dx^\mu = -c^2 dt^2 + d\vec{x}^2$$

The key idea for special relativity is difference in sign between space and time which ensures that ds^2 is invariant under Lorentz transformations (which are hyperbolic transformations),

$$dx_\nu dx^\mu = \underbrace{(\Lambda)_\nu^\alpha (\Lambda^{-1})_\beta^\mu}_{\text{Lorentz Transformations}} dx'_\alpha dx'^\beta$$

1.2 General Relativity

In order to express general relativity we need to define our mathematical objects of vectors, covectors and tensors.

1.2.1 Transforming like a Tensor

The mathematical background we need is how *vectors*, *covectors* & *tensors* transform under a coordinate change.

Vectors are objects that transform

$$V'^\mu = \frac{dx'^\mu}{dx^\alpha} V^\alpha$$

under a coordinate change $x \rightarrow x'$ they are $(1, 0)$ tensors.

Covectors are objects that transform

$$W'_\mu = \frac{dx^\mu}{dx'^\alpha} W_\alpha$$

under a coordinate change $x \rightarrow x'$ they are $(0, 1)$ tensors.

Combining these upper and lower indices we get **tensors**, transforming

$$T'^{\mu_1 \dots \mu_p}_{\nu_1 \dots \nu_q} = \frac{dx'^{\mu_1}}{dx^{\alpha_1}} \dots \frac{dx'^{\mu_p}}{dx^{\alpha_p}} \frac{dx^{\nu_1}}{dx'^{\beta_1}} \dots \frac{dx^{\nu_q}}{dx'^{\beta_q}} T^{\alpha_1 \dots \alpha_p}_{\beta_1 \dots \beta_q}$$

under a coordinate change $x \rightarrow x'$.

Tensors become the language of relativity as they transform neatly under a change of coordinates, for us this is important as once we can express physical laws in terms of tensors they'll have the same form in every reference frame.

Now we need to consider what happens when we take a derivative of our vectors or covector

$$\begin{aligned} W'_\mu &= \frac{\partial x^\alpha}{\partial x'^\mu} W_\alpha \\ \partial'_\nu W'_\mu &= \frac{\partial}{\partial x'^\nu} \left(\frac{\partial x^\alpha}{\partial x'^\mu} W_\alpha \right) \\ &= \frac{\partial}{\partial x'^\nu} \left(\frac{\partial x^\alpha}{\partial x'^\mu} \right) W_\alpha + \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial W_\alpha}{\partial x'^\nu} \\ &= \underbrace{\frac{\partial x^\beta}{\partial x'^\nu} \frac{\partial^2 x^\alpha}{\partial x'^\beta \partial x'^\mu} W_\alpha}_{\text{Not transforming like a tensor } \partial^2} + \underbrace{\frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} \partial_\beta W_\alpha}_{\text{Transforming like a (0,2 tensor)}} \end{aligned}$$

We see that we end up with another term from the chain rule complicating things. This isn't a problem in special relativity, as our transformations Λ_ν^α are linear and this second derivative disappears making vectors & covectors transform like tensors. However in general relativity we don't have that luxury and

$$\frac{\partial^2 x^\alpha}{\partial x^\beta \partial x'^\nu} \neq 0$$

To solve this we define a derivative covariant derivative of a vector and co-vector with our normal derivative and a *connection* which cancels out the extra term appearing in the chain rule,

$$D_\nu V^\mu = \partial_\nu V^\mu + \Gamma_{\nu\lambda}^\mu V^\lambda, \quad D_\nu W_\mu = \partial_\nu W_\mu + \Gamma_{\nu\mu}^\lambda W_\lambda$$

where $\Gamma_{\nu\lambda}^\mu$ is defined to transform as

$$\Gamma_{\nu\lambda}^{\prime\mu} = \frac{\partial x^\rho}{\partial x^{\prime\lambda}} \frac{\partial x^\alpha}{\partial x^{\prime\nu}} \frac{\partial x^{\prime\mu}}{\partial x^\sigma} \Gamma_{\rho\alpha}^\sigma - \frac{\partial x^\alpha}{\partial x^{\prime\nu}} \frac{\partial x^\beta}{\partial x^{\prime\lambda}} \frac{\partial^2 x^{\prime\mu}}{\partial x^\alpha \partial x^\beta} \quad (1.1)$$

This ensures derivatives transform like tensors and with a few extra conditions for convention we recover the Christoffel symbols. These are a torsion-free connection and

$$D_\mu g_{\nu\lambda} = 0$$

The Christoffel symbols are defined

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\rho} (\partial_\mu g_{\rho\nu} + \partial_\nu g_{\mu\rho} - \partial_\rho g_{\mu\nu}) \quad (1.2)$$

With the Christoffel symbols we finally can describe the Riemann tensor as a combination of these symbols $\Gamma_{\rho\alpha}^\mu \Gamma_{\lambda\nu}^\alpha$ and their derivatives $\partial_\rho \Gamma_{\lambda\nu}^\mu$ which we will need in the next section to describe curvature.

1.2.2 Curvature & metrics

Now general relativity is where we extend relativity to include mass & energy which will curve our space-time. No longer is $g_{\mu\nu}$ fixed in each component like with the Minkowski metric, we introduce a coordinate dependence $g_{\mu\nu}(x)$ which varies from point to point depending on the distribution of mass and energy which is expressed by the stress energy tensor $T_{\mu\nu}$.

The metric, and hence the curvature of our space time is governed by the the Einstein field equation,

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (1.3)$$

$G_{\mu\nu}$ is the Einstein tensor describing the curvature of spacetime expressed in terms of Ricci tensors and scalars which are contracted versions of the Riemann tensor.

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R$$

The different solutions to (1.3) give us the different spacetime geometries. The simplest is the Schwarzschild metric this is the solution for a spherical mass, where there is no electric charge or angular momentum for the mass.[2]

$$ds^2 = - \left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2. \quad (1.4)$$

This introduces the ideas of singularities, at $r = 2GM$ we see that the Schwarzschild metric diverges this will become our event horizon for when we look at black holes.

We can extend these ideas and add charge to the mass and we recover the Reissner-Nordström metric.[2]

$$ds^2 = -\Delta dt^2 + \Delta^{-1} dr^2 + r^2 d\Omega^2 \quad (1.5)$$

where,

$$\Delta = 1 - \frac{2GM}{r} + \frac{G(p^2 + q^2)}{r^2}.$$

Finally, we can take another spherical mass and allow it to rotate, this gives us the Kerr metric which we'll look at in the next section as it is the most relevant for our analysis.

2 Black Holes

Black holes are what happens when we study these metric solutions to Einstein field eq (1.3) and look at what happens at their singularities At these points the mass & energy is so dense, *singular*, that space time is so distorted that nothing not even light can escape the centre, hence the name. The boundary of our black hole can be seen in the metrics whenever they become singular, this is called the event horizons r_+ .

These objects can be described entirely just by their mass M , charge Q and angular momentum $\vec{L}$.

2.1 Schwarzschild Black Holes

The simplest black holes are the Schwarzschild black holes described by the Schwarzschild metric we talked about before (1.4), this corresponds to a spherical mass not rotating with no charge with an event horizon at $r_+ = 2GM$.

2.2 Reissner-Nordström Black Holes

This now adds charge Q to our spherical mass, they are described by the Reissner-Nordström metric discussed before (1.5). The charge modifies our event horizon, now when $\Delta = 0$ we get two singularities at

$$r_{\pm} = GM \pm \sqrt{(GM)^2 - Q^2}$$

This gives us a outer horizon r_+ and an inner *Cauchy* horizon.

2.3 Kerr Black Holes

The most relevant for our black hole thermodynamics analysis are Kerr black holes, these are Schwarzschild black holes that are rotating.[2] The Kerr metric in Boyer-Lindquist coordinates (t, r, θ, ϕ) (which reduce to Schwarzschild coordinates as $a \rightarrow 0$) is

$$ds^2 = -dt^2 + \frac{r^2}{\Delta} dr^2 + r^2 d\theta^2 + (r^2 + a^2) \sin^2 \theta d\phi^2 + \frac{2GMa}{r^2} (a \sin^2 \theta d\phi - dt)^2$$

with

$$\Delta(r) = r^2 - 2GMr + a^2$$

$$r^2(r, \theta) = r^2 + a^2 \cos^2 \theta$$

and $\vec{a} = \vec{L}/M$ represents angular momentum per mass. We get event horizons when $\Delta = 0$

$$r_{\pm} = GM \pm \sqrt{(GM)^2 - a^2}.$$

This event horizon will be where we start our black hole thermodynamics from, however before we get to that we need to discuss thermodynamics itself. [2]

3 Thermodynamics

Before we discuss the black hole analogies to thermodynamics let's briefly state the relevant laws and quantities we'll use later.

3.0.1 First Law of Thermodynamics

The first law of thermodynamics gives us the conservation of energy in thermodynamic processes. We build it from small changes in energy,

$$dE = \delta Q \pm \delta W$$

where $\pm$ depends on whether the system is doing work (-) or work is being done on it (+) and Q is the heat of the system. In reversible, closed systems this can be expressed in exact differentials rather than path dependent δ quantities. The heat added to the system is $\delta Q = TdS$ and the mechanical work $\delta W = fd\vec{X}$ with generalised coordinates or the more ubiquitous $\delta W = -PdV$

$$dE = TdS - PdV \tag{3.1}$$

$$\tag{3.2}$$

This is the form we'll try to recreate for black holes using their three quantities M , Q and $\vec{L}$.

3.0.2 Second Law of Thermodynamics

The second law of thermodynamics can be put briefly as the entropy always increases in a closed system.

$$dS \geq 0$$

Entropy is a measure of the number of microstates accessible from some macrostate, its a monotonic increasing function S maybe most elegantly described with Boltzmann entropy

$$S = k_B \ln \Omega(E)$$

where $\Omega(E)$ describes the number of microstates at a given energy. The second law tells us that systems tend towards maximising entropy which corresponds to moving away from concentrated order systems to dispersed disordered systems. It is often thought of as one of the most sacred laws in physics, here's a famous quote by astrophysicist and mathematician Arthur Eddington

“The law that entropy always increases holds, I think, the supreme position among the laws of Nature. If someone points out to you that your pet theory of the universe is in disagreement with Maxwell’s equations - then so much the worse for Maxwell’s equations. If it is found to be contradicted by observation - well, these experimentalists do bungle things sometimes. But if your theory is found to be against the Second Law of Thermodynamics I can give you no hope; there is nothing for it to collapse in deepest humiliation.” - Arthur Eddington, *New Pathways in Science*

3.1 Shannon Entropy

In information theory, Shannon entropy describes the uncertainty in a probability distribution. For a system with n states occurring with p_n is the probability, get our definition

$$S = - \sum_n p_n \ln p_n$$

In information theory there is the idea of *a bit of information*, imagine a yes-no decision. If both are equally likely

$$p_{\text{yes}} = p_{\text{no}} = \frac{1}{2}$$

we now have a system that maximises uncertainty as we have maximum ignorance on the outcome and hence we should maximise entropy.

The Shannon entropy will tell us how uncertain we are,

$$S = - \left(\frac{1}{2} \right) \ln \left(\frac{1}{2} \right) - \left(\frac{1}{2} \right) \ln \left(\frac{1}{2} \right) = \ln 2$$

This tells us that one bit of information has $\ln 2$ entropy associated with it, this will be needed when we when a particle enters into a black hole it destroys at least one bit of information, is it there or not?

4 Black Hole Thermodynamics

Now we can start to construct analogies between black hole quantities and thermodynamics, following Berkenstein. [1]

4.1 First Law

The idea around black hole thermodynamics analogy appeared from the discovery that no matter the process (mergers, accretion, etc.) the black hole horizon area increased,

$$dA \geq 0$$

Bekenstein took this observation and built on this idea mirroring the idea of the second law of thermodynamics. If area always increases just like entropy always increasing, perhaps area is entropy, and we can express black holes as thermodynamic systems. To construct this analogy we first need to construct an exact differential with at our three quantities and area to describe a black hole $dM = ...dL + ...dQ + \dots dA$ to serve as an analogue of (3.1).

4.1.1 Derivation

To save ourselves carrying a 4π around we define a *rationalized area* of a black hole as

$$\alpha = \frac{A}{4\pi}$$

Now we'll consider a Kerr black hole with mass M , charge Q and angular momentum $\vec{L}$. We can express its rational area α in terms of $\vec{a}$ the angular momentum per unit mass and the event horizon r_+ we found previously (letting $G = 1$). Where we can define $\vec{a}$ and r_+ as,

$$\vec{a} = \frac{\vec{L}}{M} \qquad r_{\pm} = M \pm \sqrt{M^2 - Q^2 - a^2}$$

Rearranging the event horizon we found we get

$$\begin{aligned} \alpha &= r_+^2 + a^2 \\ &= 2Mr_+ - Q^2 \end{aligned}$$

Now the goal is to construct an energy conservation differential like in the first law (3.1), as $E \sim M$ we'll rearrange for M .

$$M = \frac{\alpha + Q^2}{2r_+}$$

Now we need to differentiate this expression,

$$\begin{aligned} M &= d \left(\frac{\alpha + Q^2}{2r_+} \right) \\ dM &= \frac{1}{2r_+} d(\alpha + Q^2) + \frac{1}{2} (\alpha + Q^2) d(r_+^{-1}) \\ &= \frac{d\alpha}{2r_+} + \frac{2QdQ}{2r_+} - \frac{(\alpha + Q^2)}{2r_+^2} dr_+ \end{aligned}$$

Next we eliminate dr_+ terms using,

$$\begin{aligned} d(\alpha) &= d(r_+^2 + a^2) \\ d\alpha &= 2r_+ dr_+ + 2\vec{a}d\vec{a} \\ dr_+ &= \frac{d\alpha - 2\vec{a}d\vec{a}}{2r_+} \end{aligned}$$

Subbing in dr_+

$$\begin{aligned} dM &= \frac{d\alpha}{2r_+} + \frac{2QdQ}{2r_+} - \frac{\overbrace{(\alpha + Q^2)}^{2Mr_+}}{2r_+^2} \frac{d\alpha - 2\vec{a}d\vec{a}}{2r_+} \\ &= \frac{d\alpha}{2r_+} + \frac{2QdQ}{2r_+} - \frac{M}{2r_+^2} d\alpha + \frac{M}{r_+^2} \vec{a}d\vec{a} \end{aligned}$$

Now eliminating $\vec{a}$ terms

$$\begin{aligned}
&= \frac{d\alpha}{2r_+} + \frac{2QdQ}{2r_+} - \frac{M}{2r_+^2}d\alpha + \frac{M}{r_+^2} \overbrace{\vec{a}}^{=\vec{L}/M} d\vec{a} \\
&= \frac{d\alpha}{2r_+} + \frac{2QdQ}{2r_+} - \frac{M}{2r_+^2}d\alpha + \frac{1}{r_+^2}\vec{L}d\left(\frac{\vec{L}}{M}\right)
\end{aligned}$$

and differentiating

$$d\left(\frac{\vec{L}}{M}\right) = \frac{d\vec{L}}{M} - \frac{\vec{L}dM}{M^2}$$

$$\begin{aligned}
dM &= \frac{d\alpha}{2r_+} + \frac{2QdQ}{2r_+} - \frac{M}{2r_+^2}d\alpha + \frac{1}{r_+^2}\vec{L}\left(\frac{d\vec{L}}{M} - \frac{\vec{L}dM}{M^2}\right) \\
&= \left(\frac{1}{2r_+} - \frac{M}{2r_+^2}\right)d\alpha + \frac{2Q}{2r_+}dQ + \frac{\vec{L}}{Mr_+^2}d\vec{L} - \frac{L^2}{r_+^2M^2}dM
\end{aligned} \tag{4.1}$$

From line (4.1) we have everything we need we just need to tidy up the algebra,

$$\begin{aligned}
\left(1 + \frac{a^2}{r_+^2}\right)dM &= \left(\frac{1}{2r_+} - \frac{M}{2r_+^2}\right)d\alpha + \frac{2Q}{2r_+}dQ + \frac{\vec{a}}{r_+^2}d\vec{L} \\
\left(\frac{r_+^2 + a^2}{r_+^2}\right)dM &= \left(\frac{r_+ - M}{2r_+^2}\right)d\alpha + \frac{2Q}{2r_+}dQ + \frac{\vec{a}}{r_+^2}d\vec{L} \\
\left(\frac{\alpha}{r_+^2}\right)dM &= \left(\frac{r_+ - M}{2r_+^2}\right)d\alpha + \frac{2Q}{2r_+}dQ + \frac{\vec{a}}{r_+^2}d\vec{L} \\
dM &= \left(\frac{r_+ - M}{2\alpha}\right)d\alpha + \frac{2Qr_+}{2\alpha}dQ + \frac{\vec{a}}{\alpha}d\vec{L} \\
dM &= \left(\frac{r_+ - M}{2\alpha}\right)d\alpha + \Phi dQ + \vec{\Omega}d\vec{L}
\end{aligned}$$

Finally, eliminating the M term and rearranging

$$\begin{aligned}
r_+ + r_- &= M + \sqrt{M^2 - Q^2 - a^2} + M - \sqrt{M^2 - Q^2 - a^2} \\
&= 2M
\end{aligned}$$

$$\begin{aligned}
\left(\frac{r_+ - M}{2\alpha}\right) &= \frac{1}{2\alpha}(r_+ - M) \\
&= \frac{1}{2\alpha}\left(r_+ - \frac{r_+ + r_-}{2}\right) \\
&= \frac{1}{2\alpha}\left(\frac{2r_+ - r_+ - r_-}{2}\right) \\
&= \frac{1}{2\alpha}\left(\frac{r_+ - r_-}{2}\right)
\end{aligned}$$

We are left with our First Law of black hole thermodynamics,

$$dM = \frac{1}{4\alpha}(r_+ - r_-)d\alpha + \frac{Qr_+}{\alpha}dQ + \frac{\vec{a}}{M} \cdot d\vec{L}$$

Defining

$$\begin{aligned}\Theta &= \frac{1}{4\alpha}(r_+ - r_-) \\ \vec{\Omega} &= \frac{\vec{a}}{\alpha} \\ \Phi &= \frac{Qr_+}{\alpha}\end{aligned}$$

We are left with a clean analogue of the thermodynamic expression

$$dM = \Theta d\alpha + \Phi dQ + \vec{\Omega} \cdot d\vec{L} \quad (4.2)$$

Now interpreting these results we can see that ΦdQ and $\vec{\Omega} \cdot d\vec{L}$ both represent work done on the black hole. Where $\vec{\Omega}$ and Φ are the rotational angular frequency and electric potential of the black hole

$$\Phi dQ + \vec{\Omega} \cdot d\vec{L} \sim -PdV$$

That leaves us with

$$\Theta d\alpha \sim TdS$$

and we recover the correspondence with the rationalised area α and entropy S . As we said before any change of a black hole the area increases $d\alpha \geq 0$, following any change in closed thermodynamic systems $dS \geq 0$. Lastly we get our final correspondence of $\Theta \sim T$

4.2 Second Law

The first law of black hole thermodynamics tells us there is a correspondence to black hole area and black hole entropy,

$$S_{bh} \sim \alpha$$

but it doesn't tell us what it actually is. The second law aims to find the monatomic function describing black hole entropy

Applying information theory to the situation of a particle entering a black hole can give us estimates of the proportionality. When a particle falls in all the information it carries is lost to the observer, this is at least **one bit** of information which has entropy $S_{\text{one bit}} = \ln 2$. This will underpin the proportionality constant we're looking for between S_{bh} and α .

4.2.1 Derivation

We've seen from our first law derivation that there is a correspondence between $\alpha \sim S$ but what is the relationship? From here we'll refer to S_{bh} as black hole entropy to not confuse it with thermal entropy.

Lets assume that S_{bh} is a monotonically increasing function of the rationalized area,

$$S_{bh} = f(\alpha)$$

One of the simplest choices is

$$f(\alpha) = \gamma\alpha$$

for some constant γ . Now Entropy must be dimensionless so we require γ to have units of length^{-2} , this leads us to introducing $\hbar$ which has units of Planck length squared. We can then express entropy as

$$f(\alpha) = \frac{\eta}{\hbar} \alpha$$

where η is now a dimensionless constant

When a particle of rest mass μ and proper radius of b enters a black hole, the minimum possible increase in rationalised area is

$$(\Delta\alpha)_{\min} = 2\mu b$$

The radius b can not be smaller than the Compton wavelength $\frac{\hbar}{\mu}$ as a consequence of Heisenberg's uncertainty principle, so the minimum area increase is

$$(\Delta\alpha)_{\min} = 2\mu = 2\mu \left(\frac{\hbar}{\mu} \right) = 2\hbar$$

The corresponding minimum entropy increase is

$$\begin{aligned} (\Delta S_{\text{bh}})_{\min} &= \frac{df}{d\alpha} (\Delta\alpha)_{\min} \\ &= \frac{df}{d\alpha} (2\hbar) \end{aligned}$$

Now from the arguments of minimum loss of information *one bit* we let this equal $\ln 2$

$$(\Delta S_{\text{bh}})_{\min} = 2\hbar \frac{df}{d\alpha} = \ln 2$$

Solving ,

$$\begin{aligned} f(\alpha) &= \left(\frac{1}{2} \ln 2 \right) \hbar^{-1} \alpha \\ \implies S_{\text{bh}} &= \left(\frac{1}{2} \ln 2 \right) \hbar^{-1} \alpha \end{aligned} \tag{4.3}$$

4.3 Temperature

From the thermodynamic definition of temperature

$$\begin{aligned} T &= \frac{\partial E}{\partial S} \implies \frac{\partial M}{\partial S} \\ \frac{1}{T} &= \frac{\partial}{\partial M} \left(\frac{1}{2\hbar} \ln 2 \alpha \right) \\ &= \frac{1}{2\hbar} \ln 2 \frac{\partial \alpha}{\partial M} \end{aligned}$$

From the first law of black hole thermodynamics (4.2) at fixed Q and $\vec{L}$ $\partial M / \partial \alpha = \Theta$ so

$$T = \frac{2\hbar}{\ln 2} \Theta$$

This is the Bekenstein temperature of a black hole, proportional to Θ the surface gravity.

4.4 Generalised second law

A natural objection to the idea of black hole entropy appears, if some particle carries some entropy S_{object} enters the black hole the visible entropy in the universe decreases by $-S_{\text{object}}$, violating the second law.

Bekenstein resolved this idea with the generalised second law, the total entropy of the universe and black hole never decrease.

$$\Delta S_{\text{bh}} + \Delta S_{\text{universe}} \geq 0$$

Now when a particle enters a black hole $\Delta S_{\text{universe}}$ decreases but $\Delta S_{\text{bh}} \geq \Delta S_{\text{universe}}$ ensuring that the total entropy $\Delta S \geq 0$. The second law isn't violated we just need to apply it to the combined system. $\square$

5 Conclusion

Through out this reading project we aimed to show that the laws of thermodynamics have natural analogies in black hole mechanics, starting from relativistic formalisms. We studied the Schwarzschild, Reissner-Nordström and Kerr solutions of black holes, characterised by their physical properties of charge, mass and angular momentum.

After reviewing the classical first and second law of thermodynamics and defining Shannon entropy, we followed Bekenstein [1] to derive the first and second law of black hole mechanics. We then identified the parallel between the laws of thermodynamics with horizon area acting as entropy. Assigning a temperature of a black hole proportional to surface gravity, completed the analogy between the two theories. The generalised second law then extended the second law to ensure there is no violation when a particle enters by including the black hole's entropy.

References

- [1] Jacob D Bekenstein. "Black holes and entropy". In: *Physical Review D* 7.8 (1973), p. 2333.
- [2] Sean M Carroll. "Lecture notes on general relativity". In: (1997).